Complete Mathieu geometry census: manuscript and computational costs

Human Compression project

The original is Leemans–Rowley, *Rank three residually connected geometries for M_{22}, revisited*, Electronic Journal of Combinatorics 17 (2010), N4 (register L073). All nine pages are counted. The accepted replacement gives the corrected complete 81 rank-two and 431 rank-three classes, with all 431 literal source-row dispositions. It does not recover the unavailable historical `geo[]` identities. The 48-page Kılıç–Rowley predecessor supplies retained notation; it is not a second newly recovered original.

The complete written comparison. For complete raw extraction in NFC, counting characters without whitespace,

$$R_{\text{written}} = \frac{13736}{5516 + (13178 + 9480 + 13204)} = \frac{13736}{41378} = 0.331964.$$

The replacement is longer in this coordinate. All companion papers are part of the application: the ten-page class tables and nine-page source-row tables are not free outputs. The whole common paper is paid once. No earlier umbrella or another application is used without payment.

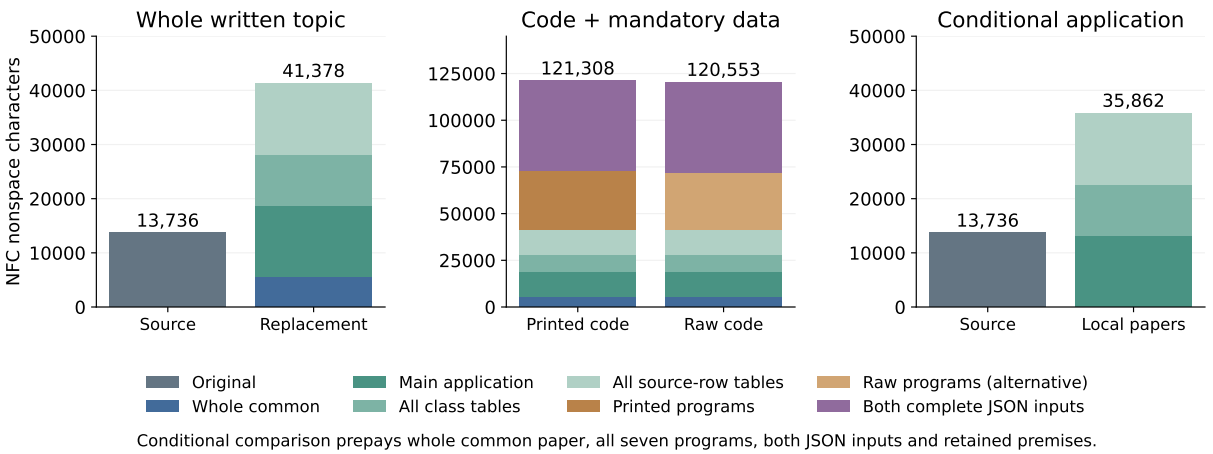
Complete document	Pages	Characters
Original L073	9	13736
Common theory U	2	5516
Main application	5	13178
All class tables	10	9480
All source-row dispositions	9	13204
Complete seven-program appendix	11	31244

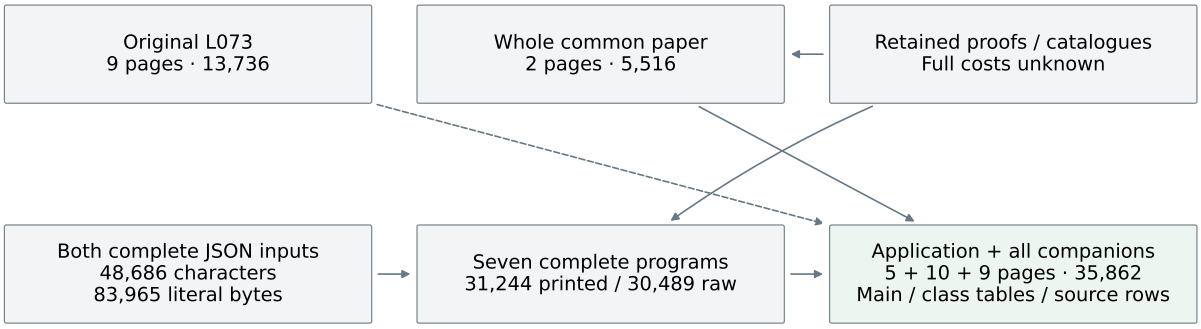
Required computation and the conditional comparison. Both computational routes pay the two complete JSON inputs: 48686 characters, or 83,965 literal file bytes. With the whole printed program appendix the separate comparison is $13736/(41378 + 31244 + 48686) = 0.113232$. Using the seven whole raw programs instead gives $13736/(41378 + 30489 + 48686) = 0.113942$. The program representations are alternatives; the JSON is required in both. There is no complete printed-data substitute or pruning of historical rows.

After acquiring *the whole common paper, all seven programs, both full JSON inputs and all retained premises*, the local application costs 35862 characters across all 24 pages:

$$r \mid (U, \text{programs}, \text{data}) = 13736/35862 = 0.383024.$$

This conditional comparison also shows an expansion; it is not a standalone whole-topic saving. The seven programs are required mathematical inputs, not merely sample diagnostics. The three auxiliary scripts and ten generated outputs remain inventoried separately, and no Lean formalization is claimed.





The graph describes complete-document acquisition, not a minimal theorem DAG. All application companions remain in every written denominator. The three auxiliary scripts contain 20,542 literal bytes (18689 nonspace characters); the ten generated outputs contain 1,567,120 bytes (838566 characters). Some outputs feed later pipeline stages; they are not charged a second time as independent external inputs. Their separate costs remain available without discount.

Unit and encoding sensitivities. Complete raw NFC coordinates are:

Unit	Written	Printed+data	Raw+data	Conditional
Whitespace tokens	0.273	0.156	0.157	0.296
Alphanumeric units	0.281	0.141	0.142	0.303
Nonspace characters	0.332	0.113	0.114	0.383
Collapsed-space characters	0.317	0.120	0.121	0.360
Collapsed-space UTF-8 bytes	0.305	0.120	0.120	0.347

All thirty coordinates and 120 exact fractions are retained in JSON/CSV: five units, NFC and NFKC, for untrimmed, furniture-selected and calibrated selected text. Selection only removes exact folios and the repeated journal footer. Six observed script/Fraktur classes account for 472 calibrated occurrences; raw text is not silently repaired. Normalized text-byte units are distinct from literal stored-file bytes.

Retained costs and scope. Named Mathieu-group identification, complete TomLib catalogue data, exact GAP/Design/GRAPE algorithms, classical design/Golay facts and predecessor notation remain explicit premises. Their full proof and catalogue-construction costs, complete graphic complexity and full formalization cost are unknown, not zero. The figures are representation lengths, not minimum proof complexity or research effort. Scientific and input acceptances are reused; this new arithmetic/reader awaits distinct review and assigns no source, question or formal credit.